

D Theoretical Analysis

In this part, we provide the theoretical results of ICL with the noisy demonstrations based on the *distinguishability* [90], which acts as a crucial condition in understanding ICL as the implicit Bayesian inference. We reveal that the noisy demonstrations can enlarge the 0-1 risk of the in-context predictor through the mismatch between the noisy prompt distribution and the clean pre-trained distributions.

First, we briefly review some definitions and preliminary setups, as well as the distinguishability condition established in [90], which is used to formalize when in-context learning occurs.

Clean and noisy demonstrations. As stated previously, the ICL techniques prompt the LLM with a few supporting examples $S_n = \{(x_i, y_i)\}_{i=1}^n$ collected in the current context, each composed of a question x_i and answer y_i , and then constructing the new input of x_{ICL} as

$$x_{\text{ICL}} = [S_n, x_{\text{test}}] = [x_1, y_1, \dots, x_n, y_n, x_{\text{test}}].$$

With S_n , the $f_\theta(x_{\text{ICL}}) \mapsto y_{\text{test}}$ is easier than $f_\theta(x_{\text{test}}) \mapsto y_{\text{test}}$. Then, the CoT further refines x_{ICL} by constructing the step-by-step *rationale* $\mathcal{T}_i$, consisting several thoughts $T_i^{(j)}$, i.e.,

$$x_{\text{CoT}} = [x_1, \mathcal{T}_1, y_1, \dots, x_n, \mathcal{T}_n, y_n, x_{\text{test}}], \quad \text{where } \mathcal{T}_i = [T_i^{(1)}, T_i^{(2)}, T_i^{(3)}, \dots, T_i^{(k)}].$$

However, as mentioned, the thoughts in CoT (Eqn. 2) can be noisy in practice. This noise can be attributed to (1) *irrelevant thoughts*, which are irrelevant but correct, or (2) *inaccurate thoughts*, which are relevant but factually wrong. Here, we uniformly formalize these two kinds of noise as

$$\hat{\mathcal{T}}_i = [T_i^{(1)}, \hat{T}_i^{(1)}, T_i^{(2)}, \hat{T}_i^{(2)}, \dots, T_i^{(k)}, \hat{T}_i^{(k)}], \quad (4)$$

where $\hat{T}_i^{(j)}$ represents a noisy thought (irrelevant or inaccurate) that is coherent with previous clean thought $T_i^{(j)}$ (relevant and correct) in Eqn. 4, and it make the x_{CoT} be noisy.

In-context learning setups. Keeping the same setups in [90], we analyze the latent concept θ from a family of concepts Θ that defines a distribution over the observed tokens o from a vocabulary $\mathcal{O}$. Following [90], we assume that the pertaining document is a length T sequence can be generated with the prior $p(\theta)$ as $p(o_1, \dots, o_T) = \int_{\theta \in \Theta} p(o_1, \dots, o_T | \theta) p(\theta) d\theta$, where $p(o_1, \dots, o_T | \theta)$ is defined by the hidden markov model. We have the prompt distribution p_{prompt} that generates the prompts for in-context learning, in which all the demonstrations share the prompt concept θ^* .

We build the derivation by analyzing the in-context predictor $f_n(x_{\text{test}}) = \arg \max_y p(y | S_n, x_{\text{test}})$, which indicates the most likely prediction over the pertaining distribution conditioned on the prompt from the prompt distribution, assuming that greedy decoding is adopted in practice. Specifically, we study the expected 0-1 error with n examples $L_{0-1}(f_n) = \mathbb{E}[\mathbb{1}[f_n(x_{\text{CoT}}) \neq y_{\text{test}}]]$.

Given the output distribution $p_\theta^i(o) := p(\mathcal{O}[i] = o | \mathcal{O}[1:i-1], \theta)$ of the i -th token with previous tokens and the analogous distribution $p_{\text{prompt}}^i := p_{\text{prompt}}(\mathcal{O}[i] = o | \mathcal{O}[1:i-1])$ under the prompt distribution, the distinguishability condition is dependent on a KL divergence between the previous two distributions and the error terms ϵ_θ resulting from the distribution mismatch between the prompt and the pertaining distributions for each example. Letting $p_\theta^i(o)$ and p_{prompt}^i correspond to the concepts θ and θ^* , the KL divergence are defined as follows,

$$\text{KL}_i(\theta^* || \theta) := \mathbb{E}_{\mathcal{O}[1:i-1] \sim p_{\text{prompt}}} [\text{KL}(p_{\text{prompt}}^i || p_\theta^i)]. \quad (5)$$

Then, the distinguishability is provided as,

Condition D.1 (distinguishability [90]). θ^* is distinguishable if for all $\theta \in \Theta, \theta \neq \theta^*$,

$$\sum_{i=1}^k \text{KL}_i(\theta^* || \theta) > \epsilon_\theta, \quad (6)$$

where ϵ_θ is a small error term corresponding to the latent concept θ . It has been empirically verified [86, 85, 89] that the larger example lengths k can increase the left-hand side and improve distinguishability. Intuitively, the larger example lengths increase the proportion of the prompt sampled from the pre-trained distribution, providing more evidence for Bayesian inference. According to the noisy demonstration introduced in Eqn. (3), we can naturally have the following condition on the distinguishability under the noisy scenario.

Condition D.2 (noisy distinguishability). With the irrelevant or inaccurate noise injection, the distribution mismatch between the prompt and the pre-trained distributions is also enlarged, resulting in a varied requirement with error term $\xi_\theta(\delta)$ (where δ indicates the noise ratio of the noisy rational in the whole demonstration) for θ^* being distinguishable if for all $\theta \in \Theta$, $\theta \neq \theta^*$,

$$\sum_{i=1}^k \text{KL}_i(\theta^* || \theta) > \epsilon_\theta + \xi_\theta(\delta), \quad \text{where} \quad \xi_\theta(\delta) \propto \delta. \quad (7)$$

Given the above Condition D.2, we can analyze the expected 0-1 error when it fails according to the noisy scenarios considered in our work. It indicates that ICL cannot induce the correct output with the noise demonstration. First, we provide the following lemma based on the results of DeVore and Lorentz [15] and Xie et al. [90], which relate the bound on KL divergence with 0-1 loss given the Condition D.2.

Lemma D.3 (noisy-relaxed bound in [90]). *let $\mathcal{B}$ denotes the set of θ which does not satisfy Condition D.2. We assume that $\text{KL}(p_{\text{prompt}}(y_{\text{test}}|x_{\text{test}})) || p(y_{\text{test}}|x_{\text{test}}, \theta)$ is bounded for all θ and θ^* minimizes the multi-class logistic risk as,*

$$L_{CE}(\theta) = -\mathbb{E}_{x_{\text{test}} \sim p_{\text{prompt}}} [p_{\text{prompt}}(y_{\text{test}}|x_{\text{test}}) \cdot \log p(y_{\text{test}}|x_{\text{test}}, \theta)]. \quad (8)$$

We can have if

$$\mathbb{E}_{x_{\text{test}} \sim p_{\text{prompt}}} [\text{KL}(p_{\text{prompt}}(y_{\text{test}}|x_{\text{test}}, \delta) || p(y_{\text{test}}|x_{\text{test}}, \theta))] \leq \epsilon_\theta + \xi_\theta(\delta), \quad \forall \quad \theta \in \mathcal{B}, \quad (9)$$

then

$$\lim_{n \rightarrow \infty} L_{0-1}(f_n) \leq \inf_f L_{0-1}(f) + g^{-1} \left(\sup_{\theta \in \mathcal{B}} (\epsilon_\theta + \xi_\theta(\delta)) \right), \quad (10)$$

where $g(\tau) = \frac{1}{2}((1-\tau)\log(1-\tau) + (1+\tau)\log(1+\tau))$ is the calibration function for the multiclass logistic loss for $\tau \in [0, 1]$.

Based on the previous work [37], we assume that there is continuity between the concept parameter θ and its corresponding output distribution p_θ , where the KL divergence is assumed to have the 2nd-order Taylor expansion. Then, we have the following theorem and proof.

Theorem D.4. *Given the previous assumption on multi-class logistic risk, we can have that $k \geq 2$ and $n \rightarrow \infty$, the risk of an in-context learning predictor f_n is bounded as*

$$\lim_{n \rightarrow \infty} L_{0-1}(f_n) \leq \inf_f L_{0-1}(f) + g^{-1} \left(O \left(\frac{\gamma_\theta \cdot \sup_{\theta \in \Theta} (\epsilon_\theta + \xi_\theta(\delta))}{k-1} \right) \right), \quad (11)$$

where $g(\tau) = \frac{1}{2}((1-\tau)\log(1-\tau) + (1+\tau)\log(1+\tau))$ is the calibration function for multi-class logistic loss for $\tau \in [0, 1]$. The overall result indicates that the upper bound of the risk is loosed by $\xi_\theta(\delta)$ of the noisy demonstration.

Proof of Theorem D.4. Assume KL divergence have a 2nd-order Taylor expansion around θ^* as,

$$\forall i > 1, \text{KL}_i(\theta^* || \theta) = \frac{1}{2}(\theta - \theta^*)^T I_{i, \theta^*} (\theta - \theta^*) + O(\|\theta - \theta^*\|^3), \quad (12)$$

where I_{i, θ^*} is the Fisher information matrix of the i -th token distribution w.r.t. θ^* . Let $\gamma_{\theta^*} = \frac{\max_i \lambda_{\max}(I_{i, \theta^*})}{\min_i \lambda_{\min}(I_{i, \theta^*})}$ and $\lambda_{\max}, \lambda_{\min}$ is the largest and smallest eigenvalues. We can have the following relationship for $\theta \in \mathcal{B}$ that,

$$\sum_{i=2}^k \text{KL}_i(\theta^* || \theta) \geq \frac{1}{2}(\theta - \theta^*)^T I_{i, \theta^*} (\theta - \theta^*) + O(\|\theta - \theta^*\|^3) \quad (13)$$

$$\geq \frac{1}{2}(k-1)\lambda_{\min}(I_{i, \theta^*})\|\theta - \theta^*\|^2, \quad (14)$$

$$\|\theta - \theta^*\|^2 \leq \frac{\epsilon_\theta + \xi_\theta(\delta)}{\frac{1}{2}(k-1)(\min_i \lambda_{\min}(I_{i, \theta^*}))}. \quad (15)$$

Then we can bound the last KL term as,

$$\text{KL}_k(\theta^* || \theta) = \frac{1}{2}(\theta - \theta^*)^T I_{i, \theta^*} (\theta - \theta^*) + O(\|\theta - \theta^*\|^3) \quad (16)$$

$$\leq \frac{1}{2}(\max_i \lambda_{\max}(I_{i, \theta^*}) \|\theta - \theta^*\|^2 + O(\|\theta - \theta^*\|^2)) \quad (17)$$

$$\leq \frac{((\epsilon_\theta + \xi_\theta(\delta)) \max_i \lambda_{\max}(I_{i, \theta^*}) + O(1))}{(k-1) \min_i \lambda_{\min}(I_{i, \theta^*})}. \quad (18)$$

Denoting $\text{KL}_k(\theta^* || \theta)$ with $\mathbb{E}_{x_{\text{test}} \sim p_{\text{prompt}}} [\text{KL}(p_{\text{prompt}}(y_{\text{test}} | x_{\text{test}}(\delta)) || p(y_{\text{test}} | x_{\text{test}}, \theta))]$, we have

$$\mathbb{E}_{x_{\text{test}} \sim p_{\text{prompt}}} [\text{KL}(p_{\text{prompt}}(y_{\text{test}} | x_{\text{test}}(\delta)) || p(y_{\text{test}} | x_{\text{test}}, \theta))] \leq \frac{((\epsilon_\theta + \xi_\theta(\delta)) \max_i \lambda_{\max}(I_{i, \theta^*}) + O(1))}{(k-1) \min_i \lambda_{\min}(I_{i, \theta^*})}. \quad (19)$$

Then, with the previous Lemma D.3, we can have the theorem result: for $k \geq 2$ and $n \rightarrow \infty$, the risk of an ICL predictor f_n is bounded as,

$$\lim_{n \rightarrow \infty} L_{0-1}(f_n) \leq \inf_f L_{0-1}(f) + g^{-1} \left(O \left(\frac{\gamma_\theta \cdot \sup_{\theta \in \Theta} (\epsilon_\theta + \xi_\theta(\delta))}{k-1} \right) \right), \quad (20)$$

where $g(\tau) = \frac{1}{2}((1-\tau) \log(1-\tau) + (1+\tau) \log(1+\tau))$ is the calibration function for multi-class logistic loss for $\tau \in [0, 1]$, and g^{-1} is roughly linear in ϵ as analyzed in [90].

□

With the above results based on [90], we can find that when the demonstration length is fixed, the multi-class logistic loss can have a loosened upper bound controlled by our noise injection. Intuitively, when the noise rate is large, the performance of ICL can be significantly affected, which can be empirically verified by the Easy-to-Hard performance trend evaluated in Tab. 3. On the other hand, considering the fixed noisy injection in the demonstration, our method can reduce the effect of the mismatching error by rephrasing the demonstration with the pre-trained distribution of the LLM itself and then decreasing the error term in RHS of Eqn. (20), which increase the distinguishability of ICL to induce the correct outputs successfully.